

GreenStar Yield Monitor and Yield Mapping System Calibration Instructions

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CALIBRATION TALK

- There are a few things customers should be aware of prior to calibrating the GreenStar system:
 - First of all, in order to achieve accurate grain weight measurement, the mass flow sensor must be calibrated at least **once per year per crop**. If the GreenStar system is not calibrated correctly, **yield and moisture readings will not be accurate**.
 - Secondly, when an operator calibrates GreenStar they are calibrating for a particular flow rate of grain through the combine. Many factors affect how crop flows through the combine and impacts the mass flow sensor. When an operator performs a standard calibration procedure, they are essentially taking a “snapshot” of a particular combination of conditions. As long as they continue to harvest at or close to these conditions, the system will maintain good accuracy. Accuracy may degrade, however, if one or more of these conditions change significantly. That is why it is important to note your ground speed and harvesting conditions while performing the calibration procedure.
 - Third, the new optional Flow Compensation calibration procedure was created to assist operators harvesting in fields where ground speed and/or yield vary considerably. While the procedure is optional, it will increase GreenStar’s accuracy when operating in these types of conditions. The principal of Flow Compensation is to measure the error at a low flow condition and then to generate a new flow compensation curve that helps correct for error in varying flow rates.
 - Fourth, prior to performing a calibration procedure, the combine must be completely empty. Upon completion of the calibration procedure, the combine must again be completely empty and all of the grain from the calibration procedure (and only that grain) must be weighed.
 - Fifth, there are two general methods of weighing grain from a calibration load:
 1. Using a grain cart with built in scales.
 2. Using truck scales.

Using a grain cart with built in scales

The scale cart method is highly recommended because it allows the weight to be determined quickly and because it involves much less handling of the grain and therefore reduces the likelihood of error. When using a scale cart to calibrate, keep the following points in mind:

1. **Calibrated** – If using a grain cart with scales, the scale on the grain cart must be calibrated correctly. GreenStar will only be as accurate as the instrument used to verify correct weight.
2. **Level** – The grain cart must be on a level surface.

3. **Zeroed** – The scale must be zeroed after the cart is stopped and just before the grain is unloaded from the combine into the cart.
4. **Stationary** – The grain cart must be stationary during the entire weighing operation. Any movement after the scale has been zeroed can introduce offsets in the weight reading. For this reason, you must stop the combine while you unload into the grain cart. You will not be able to unload on the go (UOTG) when calibrating or performing weight checks if you are using a grain cart with scales to weigh the grain.
5. If multiple combine tank loads are to be included in the calibration quantity of grain, the cart should be re-zeroed before it is filled with each additional load. The weights of the loads should be recorded on a piece of paper and totaled before being entered in the system.

Using Truck Scales

When a truck scale is used there are a number of opportunities for error associated with filling and emptying the grain carts and trucks that are used to transport the grain. If you use a truck scale to calibrate, keep the following points in mind:

1. One truck must be dedicated to weighing the calibration load. This means that no grain from any other combines may be put in that truck and that no portion of the calibration quantity of grain may be put in another truck.
2. If multiple trucks are being used to haul grain, be especially careful to use the grain weight from the correct truck for calibration.
3. The truck must be completely empty at the beginning of the calibration procedure.
4. If a grain cart is used to transfer the grain from the combine to the truck the cart must be completely empty at the beginning of the calibration procedure and it must be completely empty at the conclusion of the calibration procedure. (Note: grain carts do not empty well when filled with high moisture corn.)
5. Unloading on the go (UOTG) during calibration is difficult and discouraged unless the operator follows the calibration procedure precisely. Operators that choose to UOTG must use a truck scale and must start calibration with the combine stationary and completely empty and stop the combine on the last load so it can completely empty out.
 - Example: Operator is UOTG into a 800 bushel grain cart that empties into a semi that goes across truck scales to measure weight. The operator verifies that the combine is completely empty and goes to SETUP – Yield Monitoring or Yield Mapping – PAGE 2 and “Starts” calibration. The operator then operates the combine at its normal harvesting speed and tries to consistently run the combine at the same speed even while unloading. It takes approximately two grain cartloads to fill a semi, so on the last load the operator stops and lets the combine completely empty out. The operator

then “Stops” calibration on SETUP – Yield Monitoring or Yield Mapping – PAGE 2 and after the scale weight returns enters the correct weight.

IMPORTANT: Operators **should not UOTG** while performing Standard Calibration if they are going to perform the optional Flow Compensation calibration procedure.

IMPORTANT: UOTG increases the risk for human error while performing a calibration procedure. Only after the calibration procedure is performed correctly will the GreenStar system be accurate.

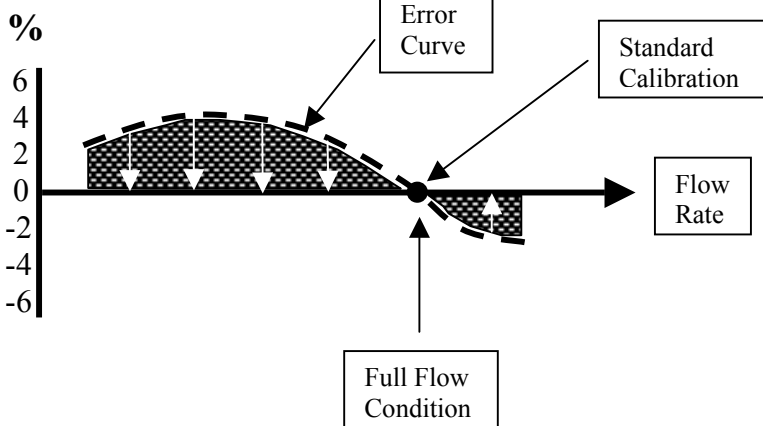
- Finally, all operators should perform the Pre-Harvest Checklist for GreenStar Yield Monitoring/Yield Mapping Systems prior to harvest. Even if the combine and/or GreenStar system is new, operators should perform this test to ensure the software is current and the hardware/components are functioning properly. The checklist can be printed from **www.StellarSupport.com** and includes checks at the combine, in the office, in the field, and on the computer with JDmap. The checklist concludes with an operation test and is an excellent way to ensure the GreenStar System is functioning properly prior to harvest.

Standard Calibration

Variations in ground speed, yield, moisture, test weight, seed size, and terrain all affect how crop flows through the combine and impacts the mass flow sensor. When an operator performs a Standard Calibration procedure, they are essentially taking a “snapshot” of a particular combination of these conditions. As long as they continue to harvest at or close to these same conditions, the system will maintain good accuracy. Accuracy may degrade, however, if one or more of these conditions change significantly.

The figure below illustrates typical errors for different grain flow rates after a Standard Calibration has been performed at the full flow condition. (The full flow condition is achieved when the combine is operated at the maximum ground speed and yield at which it is expected to operate in that crop and condition.) Percent error is the vertical distance between the flow rate axis and the dashed error curve (the area highlighted in black speckle). Notice that at the full flow condition the error is zero, but that as flow is reduced the error increases. **To compensate for varying flow rates, which can be caused by varying ground speeds and yields, Flow Compensation may need to be performed.**

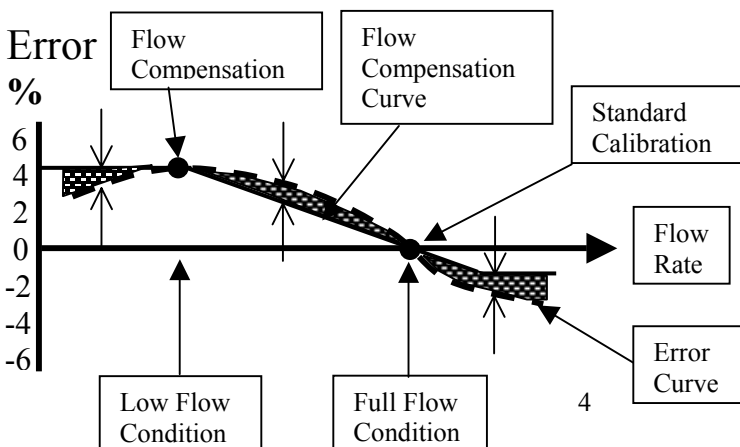
Error



Optional Flow Compensation

Flow Compensation was created to assist operators harvesting in fields where ground speed and/or yield vary considerably. (Ground speed variations typically occur in hilly or irregularly shaped fields.) While the procedure is optional, it will increase GreenStar’s accuracy when operating in these conditions. The principal of Flow Compensation is to measure the error at a low flow condition and then to generate a compensation curve that approximates the error curve (as illustrated in the previous figure) and corrects for the error.

In the figure below, a low flow condition point is established by operating the combine at half to two-thirds of the speed at which the Standard Calibration procedure was performed. By measuring error in the low flow condition, GreenStar produces a Flow Compensation curve that approximates the dashed error curve. Percent error now is the vertical distance between the flow compensation curve and the dashed error curve (the area highlighted in black speckle). Notice that this error is significantly decreased over a rather wide range of flow rates as compared with the error that occurs with Standard Calibration only.



STANDARD CALIBRATION PROCEDURE

Step	Action
1	If possible, perform calibration procedure in a field with consistent conditions and uniform yields and avoid harvesting end rows or point rows. IMPORTANT: Operators must maintain a consistent ground speed and keep the combine at full capacity. Avoid increasing and decreasing ground speed if conditions allow.
2	Operate the combine at the maximum speed you expect to drive in that crop and condition. By operating the combine at the maximum harvest speed, you are creating and calibrating for a high flow rate.
3	Decide how big a sample you want included in the calibration factor. • If performing Standard Calibration <u>only</u>, harvest at least a grain tank load. • If performing Standard Calibration followed by Low Flow Compensation procedure, harvest a grain tank load.
4	Go to SETUP – Yield Monitoring or Yield Mapping - PAGE 2.
5	Press letter “D” on the GreenStar display to select <u>Calibration Mode</u>. IMPORTANT: Make sure the Flow Comp number is set to 1.00 prior to performing Standard Calibration.
6	With the combine stopped, the combine grain tank empty, and a hauling vehicle empty, press letter “E” on the GreenStar display to “<u>Start Calibration</u>”. (Display will change to <u>Yield Calibration is Running</u>.)
7	Begin harvesting and try to keep your <u>ground speed</u> as constant as possible for the entire load. Weight displayed beside letter “F” on the GreenStar display should increase.
8	When load is complete, press letter “E” on the GreenStar display to “<u>Stop Calibration</u>” (display will change to <u>Yield Calibration is Stopped</u>) and finish with the following: • Combine grain tank <u>completely</u> empty. • All the grain from the calibration run on the hauling vehicle. • No grain from any other combine on the hauling vehicle. IMPORTANT: If you can not change Farm or Crop information after calibration, verify that “<u>Yield Calibration is Stopped</u>”.
9	Have the amount of grain in the vehicle weighed. While waiting for correct scale weight, you may continue harvesting by pressing “RUN”.

IMPORTANT: If you are using a weigh wagon or grain cart with scales to weigh the grain, make sure the wagon or grain cart has been calibrated correctly. Also, make sure the weigh wagon or grain cart is stationary, on a level surface, and zeroed prior to unloading grain.

- 10 Once you know the correct weight, go to SETUP – Yield Monitoring or Yield Mapping – PAGE 2 and press letter “F” on the GreenStar display to change the weight value.
- 11 The black arrow will toggle to “ENTER”. Using the numeric keypad, enter the correct weight.
- 12 Press letter “F” on the GreenStar display to enter the correct weight and the **Calibration Factor** will change automatically.

IMPORTANT: Calibration procedures will not change data already saved. After changes are made, all harvest information collected from that point on will reflect the changes.

NOTE: Data collected with a combine yield mapping system can be post-calibrated in JDmap.

IMPORTANT: If scale ticket weight is more than 50% higher or lower than displayed weight, the system will NOT allow entry of scale ticket weight. It is recommended that you review harvesting procedures and verify vehicle hauling grain away from combine is also following correct procedures. At that time, repeat calibration procedure.

- 13 The GreenStar system is now calibrated and should consistently be within 3 – 5% of scale ticket weight. If harvest conditions change significantly, it may be necessary to repeat Standard Calibration.

“FINE TUNNING THE CALIBRATION FACTOR”

- 14 If an operator wants to “**fine-tune**” the system further, record the new calibration factor from SETUP – Yield Monitoring or Yield Mapping – PAGE 2 on a piece of paper and repeat steps 1 – 12 at least two more times.

IMPORTANT: **The combine must be operated at approximately the same speed and under the same type of conditions the initial calibration procedure was performed.**

- 15 After each calibration procedure, record the new calibration factors on a piece of paper and then average the new calibration factors.

Example:

Load 1 new calibration factor = 856
Load 2 new calibration factor = 871
Load 3 new calibration factor = 866

New calibration factor: $856 + 871 + 866 / 3 = 864$

- 16 Enter the new averaged Calibration Factor on SETUP – Yield Monitoring or Yield Mapping – PAGE 2 by pressing letter “G”.

NOTE: By performing the procedure above and averaging the Calibration Factors, you are minimizing the variation associated with each calibration run.

Optional Flow Compensation Calibration Procedure

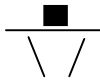
(Flow Compensation can be performed in any crop and condition.)

Step	Action
1.	During the Flow Compensation procedure, you’ll need to operate your combine at approximately one-half to two-thirds of the ground speed at which Standard Calibration procedure for this crop and condition was run and in an area that is reasonably level and uniform in yield.
2.	Harvest a 5000 – 7000 Lb. load.
3.	Go to SETUP – Yield Monitoring or Yield Mapping – PAGE 2.
4.	Press letter “D” on the GreenStar display to select <u>Low Flow Comp Mode</u> . IMPORTANT: Make sure the Flow Comp number is set to 1.00 prior to performing Flow Compensation. IMPORTANT: Message (B) with the following information may appear in Screen Section D: Low Cal Flow Comp NOT required. If this message appears, flow rate during calibration was very low. Therefore, it is neither necessary nor possible to perform optional Flow Compensation procedure. Standard Calibration procedure is sufficient.
5.	With the combine stopped, the combine grain tank empty, and a hauling vehicle empty, press letter “E” on the GreenStar display to “ <u>Start Calibration</u> ”. (Display will change to <u>Yield Calibration is Running</u> .) IMPORTANT: <u>Do not</u> unload on the go while performing Flow Compensation calibration procedure.
6.	Begin harvesting and adjust <u>ground speed</u> until moving indicator stabilizes in target range (<u>the hash marks on the GreenStar display</u>). Weight displayed beside letter “F” on the GreenStar display should increase. IMPORTANT: There is a delay after changing ground speed before moving indicator responds. Therefore, after making a ground speed adjustment, wait 8 - 10 seconds (STS) or 12 - 14 seconds (WALKER) and observe the effect of moving indicator before making another adjustment. The position of the

indicator represents flow rate. GreenStar is using ground speed to create and calibrate for a lower flow rate.

NOTE: If you are unable to consistently keep the moving indicator between the hash marks on the GreenStar display, try and maintain a consistent ground speed and keep the moving indicator consistently at or below the hash marks on the GreenStar display.

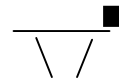
IDEAL



ACCEPTABLE



UNACCEPTABLE



7. When load is completed, press letter “E” on the GreenStar display to “**Stop Calibration**” (display will change to Yield Calibration is Stopped) and finish with the following:

- Combine grain tank **completely** empty
- All the grain from the calibration load on the hauling vehicle
- No grain from any other combine on the hauling vehicle

IMPORTANT: If you can not change Farm or Crop information after calibration, verify that “Yield Calibration is Stopped”.

NOTE: A message with the following information may appear in screen section “F” on the GreenStar display: “Comp Flow Too High. Repeat Comp Run.” If this message appears, it will not be possible to enter scale weight. Repeat Flow Compensation procedure paying special attention to keeping the moving indicator between the hash marks on the GreenStar display.

8. Have the amount of grain in the vehicle weighed. While waiting for correct scale weight, you may continue harvesting by pressing “RUN”.

IMPORTANT: If you are using a weigh wagon or grain cart with scales to weigh the grain, make sure the wagon or grain cart has been calibrated correctly. Also, make sure the weigh wagon or grain cart is stationary, on a level surface, and zeroed prior to unloading grain.

9. Once you know the correct weight, go to SETUP – Yield Monitoring or Yield Mapping – PAGE 2 and press letter “F” on the GreenStar display to change the weight value.
10. The black arrow will toggle to “ENTER”. Using the numeric keypad, enter the correct weight.
11. Press letter “F” on the GreenStar display to enter the correct weight and the **Flow Comp Number** will change automatically.

IMPORTANT: Calibration procedures will not change data already saved. After changes are made, all harvest information collected from that point on will reflect the changes.

NOTE: Data collected with a combine yield mapping system can be post-calibrated in JDmap.

IMPORTANT: If scale ticket weight is more than 50% higher or lower than displayed weight, the system will NOT allow entry of scale ticket weight. It is recommended that you review harvesting procedures and verify vehicle hauling grain away from combine is also following correct procedures. At that time, repeat calibration procedure.

12. The GreenStar system is now calibrated and should consistently be within 3 – 5% of scale ticket weight. If harvest conditions change significantly, it may be necessary to repeat both Standard Calibration and Flow Compensation calibration procedures.

“FINE TUNING THE FLOW COMP NUMBER”

13. If an operator wants to “**fine-tune**” the system further, record the new Flow Comp number from SETUP – Yield Monitoring or Yield Mapping – PAGE 2 on a piece of paper and repeat steps 1 – 12 at least two more times.

IMPORTANT: The combine must be operated at approximately the same speed and under the same type of conditions the initial Flow Comp procedure was performed.

14. After each calibration procedure, record the new Flow Comp number on a piece of paper and then average the new Flow Comp numbers.

Example:

Load 1 new Flow Comp number = .81

Load 2 new Flow Comp number = .77

Load 3 new Flow Comp number = .75

New Flow Comp Number: $.81 + .77 + .75 / 3 = .78$

15. Enter the new averaged Flow Comp number on SETUP – Yield Monitoring or Yield Mapping – PAGE 2 by pressing letter “C”.

NOTE: By performing the procedure above and averaging the Flow Comp numbers, you are minimizing the variation associated with each calibration run.

WHEN DO I NEED TO RECALIBRATE?

Prior to answering “when do I need to recalibrate?”, there are a couple of points customers need to be aware of:

- The first one is that customers must establish their own threshold (acceptable % error) for accuracy. The accuracy of the GreenStar system can be correlated with the amount of time the operator is willing to calibrate and monitor crop flow.
- Example 1: Customer A wants his or her GreenStar to be accurate but does not want to put much time into calibration or monitoring changes in crop flow. He or she goes to the field and performs a Standard Calibration Procedure and periodically checks GreenStar’s accuracy throughout the harvest season on RUN-PAGE 2. If the percent error for three consecutive loads is greater than 5% (example), he or she manually calculates a new calibration factor using the weights from all three loads and enters it on SETUP-PAGE 2. He or she continues to periodically check GreenStar’s accuracy throughout the year and only recalibrates if the percent error is greater than 5% (example).
- Example 2: Customer B wants his or her GreenStar to be as accurate as possible and is willing to put the time into calibration and monitoring changes in crop flow. He or she goes to the field and performs a Standard Calibration Procedure followed by the optional Flow Compensation Calibration Procedure and periodically checks GreenStar’s accuracy throughout the harvest season. (By performing Standard Calibration and Flow Compensation, the operator is calibrating the system at a full flow and low flow condition.) After performing both calibration procedures, he or she periodically checks GreenStar’s accuracy throughout the harvest season on RUN-PAGE 2. If the percent error is greater than 3% (example), he or she repeats standard calibration and Flow Compensation Procedures. He or she continues to periodically check GreenStar’s accuracy throughout the year and only recalibrates if the percent error is greater than 3% (example).
- The second point is that if operators want to know when to recalibrate all they have to do is check the system and compare GreenStar’s percent error to their own acceptable percent error.
 - To do this, operators should go to RUN-PAGE 2 on the GreenStar display and clear the weight and average moisture.
 - Harvest a tank load of grain.
 - When load is complete, have it weighed and compare the GreenStar weight on RUN-PAGE 2 to the actual weight.
 - If the % error is less than 5% (example), there is no need to recalibrate and the operator can resume harvesting.

Example:

Load	Scale Ticket	GreenStar	Error
1	13362	13727	2.7%

- If the % error is greater than 5% (example), repeat steps 1 – 16 of the Standard Calibration Procedure on pages 5 – 7.
- If you repeat Standard Calibration, you will need to repeat the Flow Compensation calibration procedure if you used it previously.

Note: The formula for calculating a percent error is as follows:

$$((\text{GreenStar weight} / \text{scale weight}) - 1) * 100 = \% \text{ error}$$

Note: The formula for calculating a new calibration procedure is as follows:

$$(\text{GreenStar weight} / \text{scale weight}) * \text{Old Cal. Factor} = \text{New Cal. Factor}$$